From Channel State Information to Language: Physics-Grounded Multi-Task Learning for Interpretable Wireless Channel Descriptions

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Abstract. Channel state information (CSI) is information-rich but difficult to interpret directly. We present CSI2Text, a physics-grounded multi-task framework that predicts structured propagation attributes and converts them into auditable natural-language descriptions using deterministic verbalization and relational correction. On a balanced 100k-sample UPA-8×8 dataset, the model achieves first-path delay and angle MAEs of 21.91±0.30 ns and 18.37±0.37°, while generated descriptions reach 96.43±0.12% factuality with only 0.0737±0.0078% numeric-slot hallucination. The remaining errors are concentrated in NLoS propagation: first-path angle MAE increases from 1.68° in LoS to 35.27° in NLoS. Full-model generalization tests retain 95.67±0.18% description factuality on a held-out N_f=128 configuration and 94.45±0.19% on an unseen ULA-32 array. Cross-array results further show that larger-to-smaller ULA transfer is more reliable than extrapolation to a larger unseen aperture. These findings establish structured language as a faithful interface to learned wireless-channel representations and identify robust NLoS inference as the main direction for improvement.

Introduction

CSI describes how a wireless channel transforms a transmitted signal across antennas and frequencies. It supports beam management, localization, sensing, and channel modeling, yet its complex-valued tensor form is inaccessible to most users and difficult to audit. Existing learning systems typically map CSI to a task label or a small set of regression targets. Although useful, such outputs do not provide a unified, human-readable account of the propagation scene.

We instead ask: can a model translate CSI into a factual language description of the channel? A desired output may state that a signal is indoor and NLoS, followed by the estimated first-path delay and angle, K-factor, delay and angular spreads, path richness, and reflection statistics. This formulation differs from unconstrained captioning. Wireless attributes have units, conditional semantics, and known relations; fluent text is not sufficient if the stated values contradict one another or the reference channel.

We introduce CSI2Text, a physics-grounded multi-task framework that learns a common representation for CSI, language prototypes, and propagation attributes. The CSI encoder operates on beam-frequency measurements and conditions its tokens on acquisition configuration. Three contrastive objectives align CSI, instance descriptions, and semantic prototypes. Specialized heads then model heterogeneous targets using appropriate representations: delay-profile features for temporal quantities, power-profile features for received power, and circular sine/cosine targets for angles. Finally, an explicit verbalizer renders a structured record into natural language. Because every phrase can be traced to a predicted slot, the output remains auditable.

Our contributions are:

- We formulate CSI-to-language conversion as grounded structured prediction followed by faithful verbalization, rather than unconstrained text generation.
- We develop a configuration-aware, physics-guided multi-task architecture with language alignment, shared physical features, and target-specific branches.
- We define a slot-based factuality protocol that separately measures categorical correctness, slot coverage, tolerance-aware numerical accuracy, hallucination, and physical consistency.
- We identify a clear LoS-NLoS performance asymmetry on a balanced split, showing that NLoS multipath inference - especially first-path angle and delay - is the dominant source of remaining physical error.
- We evaluate the complete multi-task model on a strictly held-out N_f=128 configuration, including physical attributes and end-to-end text factuality.
- We evaluate mixed-array learning and zero-shot array-size transfer on unseen UPA-4×4, UPA-16×4, ULA-32, and ULA-64 configurations, exposing an asymmetric ULA aperture-generalization boundary.

- We provide three-seed comparisons and ablations, plus a representative single-run relational-correction diagnostic and a reflected-path-count extension, while explicitly separating the balanced main dataset from the imbalanced extension dataset.

Related Work

Learning from CSI. Public ray-tracing resources such as DeepMIMO and RayVerse facilitate reproducible learning from channel measurements with path-level annotations. Most CSI learning pipelines optimize a single downstream task, whereas our goal is to expose several coupled propagation properties through one readable description. Physics-informed wireless modeling has recently emphasized combining physical structure with learned representations to improve interpretability and generalization .

Cross-modal and language-aligned representation learning. Contrastive language–signal alignment follows the general principle that paired modalities can share a semantic embedding space, as demonstrated at scale for images and text by CLIP . Our setting differs because CSI has no naturally authored captions and the language is derived from measurable propagation attributes. We therefore align CSI not only with instance descriptions but also with semantic prototypes and supervise the latent space using explicit physics targets.

Wireless signals and language. Recent work has connected radio-frequency measurements to language for human-centric semantic tasks. WiFi2Cap produces action captions from Wi-Fi CSI , WirelessSenseLLM aligns wireless sensing with language models for activity understanding , and RF-GPT maps RF observations to structured semantic outputs . In contrast, CSI2Text focuses on the propagation channel itself: the generated description is a factual rendering of channel geometry and statistics rather than an open-ended description of human behavior.

Method

Problem formulation

Let H in $C^{B\times F}$ denote beam-domain CSI with B beam tokens and F frequency samples, and let c contain acquisition metadata such as bandwidth and frequency configuration. Each example has a structured physical record

$$y = (e,,n_p,\tau_1,\theta_1,p_1,K,\sigma_\tau,\sigma_\theta, \tau_{LoS},\theta_{LoS},n_r,n_{rp}),$$

where e is environment, is LoS/NLoS status, n_p is path count, τ_1,θ_1,p_1 are first-path delay, angle, and power, K is the Rician K-factor, $\sigma_\tau,\sigma_\theta$ are delay and angular spreads, n_r is the total number of reflection interactions, and n_{rp} is the number of paths containing at least one reflection. The last quantity is used only in the reflected-path extension. This distinction is important: a single path may undergo several reflections, so $n_r\leq n_p$ is not a valid constraint.

The model predicts a record $y=f_(H,c)$ and a verbalizer V produces the description $s=V(y)$. Our design favors traceability: language quality is determined by the correctness and completeness of y, not by surface-form similarity alone.

Configuration-aware CSI encoder

We standardize complex CSI features per example and project each beam's frequency response into a token. The input projection combines a linear frequency projection, a depthwise–pointwise convolutional path, and adaptive average/max summaries. Configuration metadata modulates the features with a feature-wise affine transformation. Beam positional embeddings and encodings of carrier/frequency settings are then added.

A six-layer Transformer encoder with width 384, six attention heads, and a 1536-dimensional feed-forward block processes the sequence together with a learned [CLS] token. The [CLS] representation and the masked token mean are concatenated and projected to a 256-dimensional normalized CSI embedding z_c.

Language and prototype alignment

Descriptions are tokenized with a 300-token vocabulary and maximum length 48. A three-layer, four-head Transformer of width 256 produces a normalized text embedding z_t. We construct semantic prototypes from coarse channel keys that group examples by environment, LoS/NLoS state, path richness, delay and angle regimes, K-factor bin, and interaction statistics. After removing classes with fewer than five training samples, eight prototypes remain.

For a minibatch, we optimize symmetric InfoNCE losses for CSI–text, CSI–prototype, and text–prototype pairs:

$$L_{a,b} = -12N_i \leq ft[(z_a^{i}{}^{\wedge}z_b^i/T)\ _j(z_a^{i}{}^{\wedge}z_b^j/T) + (z_b^{i}{}^{\wedge}z_a^i/T)\ _j(z_b^{i}{}^{\wedge}z_a^j/T)],$$

with initial temperature $T=0.07$. A semantic classifier and a balanced K-factor attribute classifier further shape the representation.

Physics-Aware Multi-Task Learning

A residual multi-layer perceptron maps z_c to a shared physics token. It feeds an auxiliary head for normalized global targets and a set of specialized branches. Temporal targets use a delay-specific encoder and estimated power-delay-profile statistics. Power prediction uses a separate power-feature encoder. LoS and first-path angles use dedicated contexts and sine/cosine regression to respect angular periodicity. First-path delay, delay spread, first-path power, and strong-K regimes combine coarse bin classification with within-bin position regression. Reflection interactions use classification and regression heads. The path-count extension adds a distinct regressor for n_{rp}.

The total objective is

$$L=\& \lambda_{ct}L_{c,t}+ \lambda_{cp}L_{c,p}+ \lambda_{tp}L_{t,p}+ \lambda_{sem}L_{sem}+ \lambda_{attr}L_{attr} \&+_{q \in Q}\lambda_q L_q,$$

where Q contains target-specific regression, bin-classification, and within-bin losses. The three alignment losses use weight 1.0; semantic classification, attribute classification, and auxiliary regression use 0.5, 0.1, and 0.005. The principal physics weights are 0.08 for first-path delay, 0.05 each for LoS delay and LoS angle, 0.02 for first-path angle, and 0.02/0.01 for reflection classification/regression. The complete loss specification is provided in Appendix .

The implementation supports an optional relational training regularizer, but its weight is zero in all reported main runs and it is therefore not part of the reported training objective. The relational-correction results in Section instead apply constraints at the record-verbalization stage. In either setting, we deliberately do not impose the invalid relation $n_r \le n_p$.

CSI-to-Text Verbalization

The final system does not ask a large language model to invent a caption. Instead, it first predicts typed attributes and then realizes them with an explicit template. A typical output is:

This signal is indoor and NLoS, with 5 propagation paths. No direct LoS path is detected. The first-path delay is 82.4 ns, the first-path angle is -31.7°, the K-factor is 3.2 dB, and 4 paths contain reflection interactions.

Conditional clauses prevent LoS-only values from being stated for NLoS samples. This design is less linguistically diverse than open-ended generation, but it makes every statement attributable to one prediction and avoids evaluating harmless paraphrases with exact string match.

Relational Correction

Before verbalization, a bounds correction clamps impossible negative spreads and interaction counts and enforces $n_p \ge 1$. For LoS samples, a relational correction additionally replaces

$$\tau_1 \leftarrow (\tau_1,\tau_{LoS}),$$

which encodes that the detected direct path cannot arrive after the earliest propagation path. This operation changes only inconsistent records and leaves the neural predictions otherwise untouched. It is therefore a lightweight semantic safeguard rather than a second learned model.

Experimental Setup

Data. The main dataset contains 100,000 ray-traced examples with UPA-8×8 measurements and 128 frequency samples. It is balanced before splitting: 50,000 LoS and 50,000 NLoS examples. We use a fixed 80/20 train/test split. After removing semantic classes with fewer than five examples and filtering delay spread ≥400 ns, 79,429 training and 19,877 test examples remain. All main-model, baseline, and ablation comparisons use this same split.

The reflected-path extension is a separate dataset whose path labels were reconstructed and verified. It contains 100,000 examples but is naturally imbalanced: 13,628 LoS and 86,372 NLoS. Its 80,000/20,000 split contains 10,902/69,098 LoS/NLoS training examples and 2,726/17,274 test examples. We report this extension separately and do not reinterpret it as the balanced main dataset.

Full-model generalization protocols. For frequency-configuration transfer, the complete multi-task model is jointly trained on N_f in 64,96,192,256 and evaluated on $N_f=128$, which is excluded from parameter fitting. This test uses the same structured prediction and verbalization pipeline as the main model. For array transfer, a mixed-array full model is trained with UPA-8×8 and ULA-64 observations and evaluated on the seen arrays and on aligned held-out UPA-4×4, UPA-16×4, and ULA-32 observations. To test the opposite ULA size direction, a separate full model trained with ULA-32 observations is evaluated on held-out ULA-64 observations. Held-out array sets are aligned to test-group identifiers not used for fitting, preventing a newly randomized array split from reusing training links. All reported frequency and array results use seeds 0, 1, and 2 and the sample standard deviation (ddof=1).

Optimization and compute. Models are trained for 100 epochs with AdamW, batch size 128, initial learning rate 3×10^{-4}, weight decay 10^{-2}, five epochs of linear warmup, and cosine decay to 10^{-5}. Checkpoints are saved every 20 epochs. Unless noted otherwise, results are the mean and sample standard deviation over seeds 0, 1, and 2. The relational training-regularizer weight is zero for these runs; relational correction is applied only during record verbalization in the diagnostic reported in Section . The model used for the balanced three-seed main experiments has 28,938,178 trainable parameters (28.94M), as recorded independently in all three training logs. Each run uses one NVIDIA GeForce RTX 4090 with 23.6 GB memory; four such GPUs were available for parallel experiments. The software stack is PyTorch 2.5.1+cu121 and CUDA 12.1. Automatic mixed precision is disabled. A representative run takes 5 h 27 min 44 s.

Metrics. For physical regression we report MAE; for reflection interactions we additionally report exact and adjacent-bin accuracy. Text evaluation parses predicted and target descriptions into typed slots. Categorical fields are measured by accuracy and macro-F1. Numeric slot F1 evaluates whether the required values are present, while hallucination is the fraction of predicted numeric slots unsupported by the reference schema. Tolerance accuracy measures whether a present predicted value lies within a field-specific tolerance: 50 ns for delays, 15° for angles, 5 dB for first-path power, 3 dB for K, and one count for path/interaction fields. The reported composite description factuality is the macro average of categorical accuracies, numeric-slot F1, numeric tolerance accuracy, and 1-hallucination. It is not sentence-level exact match. Physical consistency is an auxiliary rule-based diagnostic applied to the predicted record.

Baseline Comparison

We compare with a flattened-CSI MLP, a convolutional baseline, an inverse-FFT power-delay-profile (PDP/IFFT) MLP, a Transformer without specialized physics branches, and the same CSI encoder trained with independent single-task heads. Table reports the common three-seed evaluation.

Physical prediction on the balanced main split (mean ± sample standard deviation over seeds 0, 1, and 2). Lower is better for MAE; higher is better for accuracy. ``Refl.'' denotes total reflection interactions, not reflected paths. Best results are bold.

Method	τ_1 MAE (ns) ↓	K MAE (dB) ↓	Refl. MAE ↓	Exact ↑	Adjacent ↑	θ_1 MAE (°) ↓	p_1 MAE (dB) ↓
CSI2Text full multi-task	21.911±0.303	1.983±0.263	0.983±0.005	0.7061±0.0039	0.9742±0.0007	18.372±0.365	4.958±0.269
CSI encoder + single-task head	51.164±6.248	1.521±0.090	1.222±0.072	0.5523±0.0259	0.9594±0.0126	20.449±0.431	3.740±0.209
PDP/IFFT + MLP	28.750±4.611	5.002±0.275	1.706±0.032	0.4420±0.0201	0.9055±0.0019	40.292±0.159	12.236±0.278
Flattened CSI + MLP	114.371±0.978	5.600±1.013	1.835±0.196	0.4237±0.0999	0.8930±0.0431	32.782±0.413	13.315±1.221
Transformer without branches	268.177±0.003	22.762±0.000	3.214±0.021	0.1808±0.0000	0.5359±0.0000	53.111±0.005	38.090±0.013
CNN baseline	1608.970±1202.800	56.869±34.742	119.488±185.779	0.1562±0.1463	0.3107±0.1804	102.231±42.561	14591.900±18581.100

The full model is best on five of seven metrics, including the propagation quantities most dependent on coupled temporal and angular structure. The single-task system remains stronger for K-factor and first-path power, showing that multi-task sharing is not uniformly beneficial. Compared with PDP/IFFT+MLP, the full system reduces first-path delay MAE by 23.8% while also improving angle and reflection predictions.

LoS versus NLoS stratified performance

Propagation-regime-stratified MAE on the balanced main split (mean ± sample standard deviation over three seeds). Lower is better. The ratio is NLoS MAE divided by LoS MAE.

Target	LoS MAE	NLoS MAE	Ratio
First-path delay	8.174±0.156 ns	35.820±0.599 ns	4.38×
First-path angle	1.680±0.043°	35.272±0.748°	21.00×
First-path power	2.973±0.638 dB	6.966±0.122 dB	2.34×

Table exposes the central limitation of the current model. LoS predictions are highly accurate, particularly for first-path angle, whose MAE is only 1.68°. When the direct path is blocked, delay error increases by 4.38×, angle error by 21.00×, and power error by 2.34×. Because the main dataset is balanced before splitting, this gap cannot be attributed to a shortage of NLoS training examples; it instead reflects the intrinsic ambiguity and richer multipath structure of NLoS propagation. Accordingly, improvements to NLoS representation learning are the primary technical priority.

Full-Model Cross-Configuration Generalization

Held-out frequency configuration

We first hold out N_f=128 while jointly training the complete multi-task model on N_f in 64,96,192,256 . This experiment evaluates the same full prediction and verbalization pipeline used elsewhere in the paper. Table shows stable three-seed transfer to the unseen frequency-sampling configuration.

Full-model results on the held-out N_f=128 configuration (mean ± sample standard deviation over seeds 0, 1, and 2).

Metric	Held-out N_f=128
LoS/NLoS accuracy ↑	99.827±0.006%
First-path delay MAE ↓	58.752±1.338 ns
LoS delay MAE ↓	21.801±1.270 ns
K-factor MAE ↓	2.166±0.037 dB
First-path angle MAE ↓	20.483±0.146°
First-path power MAE ↓	5.998±0.355 dB
Delay-spread MAE ↓	19.604±0.253 ns
Azimuth-spread MAE ↓	10.952±1.399°
Reflection-interaction exact accuracy ↑	68.303±0.268%
Reflected-path exact accuracy ↑	95.547±0.064%

The aggregate result again masks a large propagation-regime gap. For first-path delay, LoS and NLoS MAEs are 25.123±0.419 and 92.393±2.271 ns; for first-path angle they are 1.756±0.197° and 39.215±0.279°. First-path power is less asymmetric but remains harder in NLoS (5.079±0.507 versus 6.918±0.226 dB). Thus, the excellent LoS geometry transfer and the persistent NLoS bottleneck observed on the main split also hold under an unseen N_f.

The held-out-N_f descriptions are generated end to end, without oracle values for non-delay fields. They achieve 95.667±0.179% composite factuality, 99.883±0.008% categorical macro-F1, 99.977±0.002% numeric-slot F1, 78.639±0.906% numerical tolerance accuracy, and only 0.0455±0.0038% numeric-slot hallucination. Post-verbalization physical consistency is 99.122±0.215% after relational correction. These results support full-model transfer to the held-out N_f=128 configuration, but do not imply uniform zero-shot performance for arbitrary sampling densities or carrier frequencies.

Cross-array and array-size transfer

Table evaluates the mixed UPA/ULA full model on seen and held-out array sizes. The unseen arrays use aligned held-out link groups rather than newly randomized splits. This distinction makes the comparison a test of array configuration transfer rather than memorization of the same physical links.

Full-model cross-array results (mean ± sample standard deviation over three seeds). UPA-8×8 and ULA-64 are seen during mixed-array training; the remaining configurations are held out.

Array	Status	LoS angle MAE	NLoS first-angle MAE	First-delay MAE	K MAE	Text factuality
UPA-8×8	Seen	6.106±1.966°	42.591±0.377°	53.854±6.213 ns	4.173±0.313 dB	94.316±0.096%
ULA-64	Seen	14.566±2.623°	42.710±0.562°	47.449±2.121 ns	4.310±0.249 dB	94.178±0.167%
UPA-4×4	Unseen	10.257±3.013°	45.859±0.110°	44.082±0.786 ns	5.884±0.230 dB	93.556±0.092%
UPA-16×4	Unseen	9.446±0.913°	43.237±0.114°	54.828±5.982 ns	4.758±0.457 dB	93.639±0.136%
ULA-32	Unseen	11.345±0.566°	42.919±0.400°	43.399±1.029 ns	4.282±0.285 dB	94.449±0.193%

The unseen UPA sizes preserve high categorical and textual reliability, while their LoS-angle MAEs remain within 9.45–10.26°. The unseen ULA-32 result is similarly stable: LoS/NLoS accuracy is 97.037±0.220%, reflected-path exact accuracy is 94.340±0.165%, and numerical tolerance accuracy is 75.881±0.957%. Across every array in Table , however, NLoS first-path angle remains approximately 42–46°. Its weak dependence on array type or size indicates a shared NLoS inference limitation rather than a failure specific to one array geometry.

We additionally reverse the ULA size direction in a separately trained full model. Table shows that ULA-64-to-ULA-32 transfer is substantially easier than ULA-32-to-ULA-64 transfer.

Directional ULA size transfer. Each row reports the seen training size and its unseen test size over three seeds.

Direction	Status	LoS angle	First delay	Text factuality
ULA-64 → ULA-32	Seen 64	14.566±2.623°	47.449±2.121 ns	94.178±0.167%
	Unseen 32	11.345±0.566°	43.399±1.029 ns	94.449±0.193%
ULA-32 → ULA-64	Seen 32	10.730±0.785°	41.619±0.954 ns	95.092±0.096%
	Unseen 64	17.223±3.331°	49.350±3.524 ns	94.222±0.073%

The larger-to-smaller direction exhibits no measurable degradation on these aggregate metrics, whereas the smaller-to-larger direction increases LoS-angle MAE by 6.49° and first-path-delay MAE by 7.73 ns. We do not interpret the slightly better unseen ULA-32 numbers as an intrinsic advantage of zero-shot prediction; they indicate that the smaller aperture lies within the representation learned from the larger one. Conversely, extrapolating to a larger unseen aperture changes the spatial phase structure and angular resolution, making it the harder direction.

Three-Seed Text Factuality Results

Description quality on the balanced main split (mean ± sample standard deviation over seeds 0, 1, and 2).

Metric	Result
Composite description factuality ↑	0.9643 ± 0.0012
Numeric-slot F1 ↑	0.99959 ± 0.00003
Numeric tolerance accuracy ↑	0.8261 ± 0.0058
Numeric-slot hallucination ↓	0.000737 ± 0.000078

Table shows that, on the balanced main split, the system almost always emits the required slots and rarely introduces an unsupported one. The three seed-wise numerical tolerance accuracies are 81.97%, 82.78%, and 83.10%, indicating substantially less seed sensitivity than in the imbalanced reflected-path extension. This metric is computed over jointly present numerical fields; it is not the fraction of entire sentences that are exactly correct.

Field-level results in Appendix show that LoS delay, LoS angle, delay spread, path count, and first-path delay are the most reliable numerical statements. The principal remaining errors are first-path power (64.36±4.30% within 5 dB), reflection-interaction count (65.62±0.31% within one), and first-path angle (68.41±1.02% within 15°). Angle-spread accuracy has the largest cross-seed variation (79.07±7.16%). The rule-based physical-consistency diagnostic is not used as a primary factuality measure because it depends on the chosen rule set and post-processing; factual claims are evaluated directly against target slots.

Reflected-Path Extension

The base model predicts the total number of reflection interactions. To support statements such as ``four paths contain reflection interactions,'' we add a separate reflected-path-count head. This extension is strongly NLoS-skewed and is not used for the balanced main comparison. On its separate test split, the verbalized reflected-path value attains an MAE of 0.091±0.007, RMSE of 0.234±0.008, and accuracy within one path of 98.82±0.06%. These are text-level metrics computed after record generation and verbalization; they should not be confused with head-level exact classification accuracy. Across the complete extension descriptions, composite factuality is 94.06±1.07%, numeric-slot F1 is 99.974±0.012%, numeric-slot hallucination is 0.0437±0.0200%, numerical tolerance accuracy is 70.63±5.48%, and physical consistency is 99.872±0.029%. The strong result demonstrates that the two reflection concepts can be modeled separately without relying on the invalid assumption that total interactions cannot exceed total paths.

Ablation and Error Analysis

Architectural ablations. Table compares the full model with removal of the shared physics token or delay-specific encoder. Removing the delay encoder degrades first-path delay, delay spread, K-factor, power, and reflection prediction, confirming the value of target-aware temporal features. The shared token has a mixed effect: it improves LoS angle and reflection exact accuracy, but the no-shared variant is slightly better on several regression metrics. We therefore interpret it as a useful coupling mechanism, not a universally superior component.

Selected three-seed ablations. Lower is better except reflection exact accuracy.

Model	τ_1 (ns)	σ_τ (ns)	K (dB)	Refl. exact
Full	21.91±0.30	11.17±0.11	1.98±0.26	0.7061±0.0039
No shared physics token	21.62±0.71	11.15±0.08	1.74±0.04	0.7013±0.0041
No delay-specific encoder	69.02±1.46	29.64±0.84	2.54±0.12	0.6652±0.0057

Effect of relational correction. In a representative run, bounds-only correction leaves 1,426 cases in which the predicted first path precedes the predicted LoS path. Adding the relational correction reduces this count to zero and raises physical consistency from 92.64% to 99.86%. Composite factuality changes from 95.6068% to 95.6051%, and numerical tolerance accuracy from 78.4292% to 78.4205%. Thus, the rule removes contradictions at negligible measured accuracy cost. We treat this consistency score as a diagnostic, not a substitute for comparison with ground-truth slots.

Failure modes. The dominant failure mode is continuous-value estimation under NLoS propagation rather than slot omission or category recognition. As Table shows, the model is already accurate for key LoS quantities but degrades sharply when the direct path is absent, most notably for first-path angle and delay. Angle spread and K-factor also have cross-seed variance, and the NLoS-skewed path-count extension further stresses this difficult regime. Future work should therefore prioritize NLoS-aware objectives, multipath-structure encoders, hard-example reweighting, and uncertainty calibration, while continuing to report LoS/NLoS-stratified errors instead of relying only on aggregate MAE.

Limitations

First, the output language is template-based and therefore factual and auditable but not stylistically open-ended. The work demonstrates CSI-to-natural-language verbalization, not unrestricted dialogue about wireless scenes. Second, the frequency experiment holds out one sampling configuration, N_f=128; it does not establish uniform transfer to arbitrary N_f, bandwidths, subcarrier spacings, or carrier frequencies. Third, the array experiments cover a limited set of simulated UPA and ULA geometries. They demonstrate transfer to unseen array sizes, but smaller-to-larger ULA extrapolation remains weaker and the results do not yet establish robustness to element perturbations, calibration errors, noise, or measured channels. Fourth, multi-task learning does not dominate single-task learning for every variable, especially K-factor and first-path power. Fifth, tolerance-based numerical accuracy depends on application-specific thresholds and should always be presented with raw error metrics. Sixth, the large LoS–NLoS gap persists under both unseen frequency and unseen array configurations; improving representation of indirect and higher-order multipath components is therefore the main modeling limitation. Finally, the reflected-path extension is strongly NLoS-skewed, so its aggregate score should not be compared directly with balanced main-split results.

Conclusion

We presented a physics-grounded framework that converts CSI into an interpretable natural-language channel description. The system combines configuration-aware CSI encoding, language and prototype alignment, specialized multi-task physics heads, explicit verbalization, and a lightweight relational correction. On the balanced main split, three-seed experiments show strong physical prediction, near-complete slot realization, and very low hallucination. The most important empirical finding is the propagation-regime asymmetry: the model is highly accurate in LoS conditions, while NLoS first-path delay, angle, and power remain substantially harder. A representative single-run diagnostic shows that relational verbalization correction removes contradictions with negligible factuality change. Separately, the NLoS-skewed reflected-path extension distinguishes reflection interactions from the number of reflected paths and is not included in the balanced main comparison. The complete multi-task model also transfers to a held-out $N_f=128$ configuration, achieving 58.75 ± 1.34 ns first-path-delay MAE, $99.827\pm0.006\%$ LoS/NLoS accuracy, and $95.67\pm0.18\%$ end-to-end description factuality. Mixed-array training further supports zero-shot size transfer to UPA-4×4, UPA-16×4, and ULA-32, while a reverse ULA experiment shows that extrapolating from ULA-32 to unseen ULA-64 is harder than transferring from ULA-64 to ULA-32. Across these configuration shifts, LoS geometry remains substantially more reliable than NLoS first-path inference, which remains the dominant generalization bottleneck. The results support structured language as a practical interface for inspecting learned wireless channel models, while identifying NLoS-specific representation learning and cross-configuration validation as the next priorities.

Reproducibility statement

The anonymous supplementary code includes preprocessing, training, evaluation, signal-description parsing, and metric aggregation scripts. Appendix records the optimization and loss configuration, and Section reports data filtering, seeds, compute, and software versions.

References

Radford, Alec and Kim, Jong Wook and Hallacy, Chris and Ramesh, Aditya and Goh, Gabriel and Agarwal, Sandhini and Sastry, Girish and Askell, Amanda and Mishkin, Pamela and Clark, Jack and Krueger, Gretchen and Sutskever, Ilya. Learning Transferable Visual Models From Natural Language Supervision. Proceedings of the 38th International Conference on Machine Learning. 2021.

Vaswani, Ashish and Shazeer, Noam and Parmar, Niki and Uszkoreit, Jakob and Jones, Llion and Gomez, Aidan N. and Kaiser, Lukasz and Polosukhin, Illia. Attention Is All You Need. Advances in Neural Information Processing Systems. 2017.

Alkhateeb, Ahmed. DeepMIMO: A Generic Deep Learning Dataset for Millimeter Wave and Massive MIMO Applications. arXiv preprint arXiv:1902.06435. 2019.

UNIC Lab. RayVerse: A Large-Scale Urban Radio-Propagation Dataset Generated Using Ray Tracing. 2025.

Zhu, Guangxu and Sun, Yuanming and Ji, Kaibin. Physics-Informed Generalizable Wireless Channel Modeling with Segmentation and Deep Learning: Fundamentals, Methodologies, and Challenges. arXiv preprint arXiv:2401.01288. 2024.

Wei, Tzu-Ti and Huang, Chu-Yu and Tseng, Yu-Chee and Chen, Jen-Jee. WiFi2Cap: Semantic Action Captioning from Wi-Fi CSI via Limb-Level Semantic Alignment. arXiv preprint arXiv:2603.22690. 2026.

Keya, Mahmuda and Pillai, Sneh and Yuan, Jiawei and Zeng, Kai and Jiao, Long. WirelessSenseLLM: Zero-Shot Human Activity Understanding by Bridging Wireless Signals and Human Language. arXiv preprint arXiv:2605.14070. 2026.

Zou, Hang and Tian, Yu and Wang, Bohao and Bariah, Lina and Lasaulce, Samson and Huang, Chongwen and Debbah, M 'erouane. RF-GPT: Teaching AI to See the Wireless World. arXiv preprint arXiv:2602.14833. 2026.

Loss weights

Loss configuration used for the main full model.

Group	Term	Weight
Alignment	CSI–text / CSI–prototype / text–prototype	1.0 / 1.0 / 1.0
Representation	semantic classifier / attribute classifier / auxiliary physics regression	0.5 / 0.1 / 0.005
K-factor	strong-bin classifier / within-bin position	0.5 / 0.1
First-path delay	normalized / raw / fused raw	0.08 / 0.001 / 0.02
First-path delay	bin classifier / position / consistency / tail underestimate	0.02 / 0.01 / 0.01 / 0.01
Delay spread	raw / bin classifier / within-bin position	0.0005 / 0.03 / 0.008
LoS delay	regression / nonnegative / consistency	0.05 / 0.005 / 0.005
Angles	LoS / first path / first-path NLoS	0.05 / 0.02 / 0.01
First-path power	bin classifier / position; NLoS sample multiplier	0.03 / 0.1; 2.0
Reflection interactions	classifier / regression; NLoS sample multiplier	0.02 / 0.01; 1.0

Field-specific text metrics

Three-seed numerical tolerance accuracy on the balanced main split. Reflected-path count is excluded from this table and reported only in the separate path-count extension.

Field	Tolerance accuracy
Path count (±1)	0.9533±0.0061
First-path delay (±50 ns)	0.9451±0.0032
First-path angle (±15°)	0.6841±0.0102
First-path power (±5 dB)	0.6436±0.0430
K-factor (±3 dB)	0.7971±0.0302
Delay spread (±50 ns)	0.9700±0.0003
Angle spread (±15°)	0.7907±0.0716
LoS delay (±50 ns)	0.9915±0.0012
LoS angle (±15°)	0.9972±0.0007
Reflection interactions (±1)	0.6562±0.0031